

Economic growth Notebook

[Code ▼](#)

We will use a dataset that was used in the paper: I just ran two million regressions by Sala-I-Martin and Model uncertainty in cross country growth regression by Fernandez et. al. The dataset has 41 possible explanatory variables with 72 countries. The data consists of 43 columns with the Country, y (economic growth in per capita GDP) and 41 possible predictor variables.

[Hide](#)

```
rm(list=ls())  
eg <- read.csv("economicgrowth.csv")  
str(eg)
```

```

'data.frame': 72 obs. of 43 variables:
 $ Country      : chr  "DZ" "AR" "AU" "AT" ...
 $ y            : num  0.01369 0.00642 0.0189 0.02922 0.02857 ...
 $ Abslat       : num  36.7 36.7 32.2 48.2 50.8 ...
 $ Spanish      : int  0 1 0 0 0 1 0 0 0 0 ...
 $ French       : int  1 0 0 0 0 0 0 0 1 0 ...
 $ Brit         : int  0 0 1 0 0 0 1 0 0 1 ...
 $ WarDummy     : int  0 1 0 0 0 0 0 0 0 0 ...
 $ LatAmerica   : int  0 1 0 0 0 1 0 1 0 0 ...
 $ SubSahara    : int  0 0 0 0 0 0 1 0 1 0 ...
 $ OutwarOr     : int  1 0 0 0 1 0 0 0 1 1 ...
 $ Area         : int  2382 2767 7687 84 31 1099 582 8512 475 9976 ...
 $ PrScEnroll   : num  0.46 0.98 1 1 1 0.64 0.42 0.95 0.65 1 ...
 $ LifeExp      : num  47.3 65.2 70.7 68.8 69.7 42.8 45.7 54.9 43.4 71.1 ...
 $ GDP60       : num  7.44 8.1 8.97 8.55 8.62 ...
 $ Mining       : num  0.196 0.023 0.038 0.004 0 0.075 0.533 0.017 0.088 0.034 ...
 $ EcoOrg       : int  0 5 5 4 5 3 5 3 5 5 ...
 $ YrsOpen      : num  0 0.089 0.689 0.778 0.778 0.733 0.356 0.089 0.156 0.956 ...
 $ Age         : int  84 0 0 0 88 0 0 0 0 0 ...
 $ Buddha       : num  0 0 0 0 0 0 0 0 0 0 ...
 $ Catholic     : num  0 0.9 0.26 0.85 0.75 0.95 0.25 0.9 0.16 0.46 ...
 $ Confucian    : num  0 0 0 0 0 0 0 0 0 0 ...
 $ EthnoL       : num  0.43 0.31 0.32 0.13 0.55 0.68 0.51 0.07 0.89 0.75 ...
 $ Hindu        : num  0 0 0 0 0 0 0 0 0 0 ...
 $ Jewish       : num  0.005 0.02 0 0 0 0 0 0 0 0 ...
 $ Muslim       : num  0.99 0 0 0 0 0 0 0 0.16 0 ...
 $ PrExports    : num  0.933 0.861 0.811 0.198 0.238 0.969 0.75 0.858 0.915 0.488 ...
 $ Protestants : num  0.005 0.02 0.5 0.06 0.13 0.03 0.25 0 0.17 0.26 ...
 $ RuleofLaw    : num  0.333 0.333 1 1 1 ...
 $ Popg         : num  0.02842 0.01525 0.01645 0.00237 0.00249 ...
 $ WorkPop      : num  -1.33 -0.933 -0.902 -0.734 -0.954 ...
 $ LabForce     : num  2856 8110 4186 3384 3526 ...
 $ HighEnroll   : num  0.003 0.109 0.131 0.08 0.09 0.036 0 0.016 0.002 0.16 ...
 $ PubIEdupct   : num  0.0297 0.0223 0.0217 0.0248 0.0463 0.0158 0.03 0.0041 0.0221 0.0365 ...
 $ RevnCoup     : num  0.123 0.96 0 0 0 1.19 0 0.12 0.08 0 ...
 $ PolRights    : num  5.83 3.94 1 1 1 ...
 $ CivlLib      : num  5.89 3.56 1 1 1 ...
 $ English      : num  0 0 0.95 0 0 0 0 0 0 0.614 ...
 $ Foreign      : num  0.836 0.836 0 0.98 0.345 0.372 0 0.998 0.007 0.243 ...
 $ RFEXDist     : int  190 113 129 100 100 181 131 97 187 83 ...
 $ EquipInv     : num  0.0507 0.016 0.095 0.1006 0.0635 ...
 $ NequipInv    : num  0.191 0.12 0.221 0.134 0.171 ...
 $ stdBMP       : num  186.555 66.85 0.001 0.001 0.001 ...
 $ BlMktPm      : num  0.131 0.16 0 0 0 0.03 0.072 0.24 0.05 0 ...

```

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```

eg1 <- subset(eg,select=-c(Country))
#Alternatively eg1 <- eg[,2:43]
str(eg1)

```

```

'data.frame':  72 obs. of  42 variables:
 $ y          : num  0.01369 0.00642 0.0189 0.02922 0.02857 ...
 $ Abslat     : num  36.7 36.7 32.2 48.2 50.8 ...
 $ Spanish    : int   0 1 0 0 0 1 0 0 0 0 ...
 $ French     : int   1 0 0 0 0 0 0 0 1 0 ...
 $ Brit       : int   0 0 1 0 0 0 1 0 0 1 ...
 $ WarDummy   : int   0 1 0 0 0 0 0 0 0 0 ...
 $ LatAmerica : int   0 1 0 0 0 1 0 1 0 0 ...
 $ SubSahara  : int   0 0 0 0 0 0 1 0 1 0 ...
 $ OutwarOr   : int   1 0 0 0 1 0 0 0 1 1 ...
 $ Area       : int  2382 2767 7687 84 31 1099 582 8512 475 9976 ...
 $ PrScEnroll : num   0.46 0.98 1 1 1 0.64 0.42 0.95 0.65 1 ...
 $ LifeExp    : num  47.3 65.2 70.7 68.8 69.7 42.8 45.7 54.9 43.4 71.1 ...
 $ GDP60      : num   7.44 8.1 8.97 8.55 8.62 ...
 $ Mining     : num   0.196 0.023 0.038 0.004 0 0.075 0.533 0.017 0.088 0.034 ...
 $ EcoOrg     : int   0 5 5 4 5 3 5 3 5 5 ...
 $ YrsOpen    : num   0 0.089 0.689 0.778 0.778 0.733 0.356 0.089 0.156 0.956 ...
 $ Age        : int  84 0 0 0 88 0 0 0 0 0 ...
 $ Buddha     : num   0 0 0 0 0 0 0 0 0 0 ...
 $ Catholic   : num   0 0.9 0.26 0.85 0.75 0.95 0.25 0.9 0.16 0.46 ...
 $ Confucian  : num   0 0 0 0 0 0 0 0 0 0 ...
 $ EthnoL     : num   0.43 0.31 0.32 0.13 0.55 0.68 0.51 0.07 0.89 0.75 ...
 $ Hindu      : num   0 0 0 0 0 0 0 0 0 0 ...
 $ Jewish     : num   0.005 0.02 0 0 0 0 0 0 0 0 ...
 $ Muslim     : num   0.99 0 0 0 0 0 0 0 0.16 0 ...
 $ PrExports  : num   0.933 0.861 0.811 0.198 0.238 0.969 0.75 0.858 0.915 0.488 ...
 $ Protestants: num   0.005 0.02 0.5 0.06 0.13 0.03 0.25 0 0.17 0.26 ...
 $ RuleofLaw  : num   0.333 0.333 1 1 1 ...
 $ Popg       : num   0.02842 0.01525 0.01645 0.00237 0.00249 ...
 $ WorkPop    : num  -1.33 -0.933 -0.902 -0.734 -0.954 ...
 $ LabForce   : num  2856 8110 4186 3384 3526 ...
 $ HighEnroll : num   0.003 0.109 0.131 0.08 0.09 0.036 0 0.016 0.002 0.16 ...
 $ PubLEdupct : num   0.0297 0.0223 0.0217 0.0248 0.0463 0.0158 0.03 0.0041 0.0221 0.0365 ...
 $ RevnCoup   : num   0.123 0.96 0 0 0 1.19 0 0.12 0.08 0 ...
 $ PolRights  : num   5.83 3.94 1 1 1 ...
 $ CivlLib    : num   5.89 3.56 1 1 1 ...
 $ English    : num   0 0 0.95 0 0 0 0 0 0 0.614 ...
 $ Foreign    : num   0.836 0.836 0 0.98 0.345 0.372 0 0.998 0.007 0.243 ...
 $ RFEXDist   : int  190 113 129 100 100 181 131 97 187 83 ...
 $ EquipInv   : num   0.0507 0.016 0.095 0.1006 0.0635 ...
 $ NequipInv  : num   0.191 0.12 0.221 0.134 0.171 ...
 $ stdBMP     : num  186.555 66.85 0.001 0.001 0.001 ...
 $ BLMktPm    : num   0.131 0.16 0 0 0 0.03 0.072 0.24 0.05 0 ...

```

Here if we use all 41 variables to explain y , it is reasonable to expect a good fit ($p = 41$, $n = 72$). We are interested in finding smaller sized subsets of variables (few variables) that provide good fit to the data.

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```

model0 <- lm(y~.,data=eg1)
summary(model0)

```

Call:

```
lm(formula = y ~ ., data = eg1)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-0.0090519	-0.0023153	0.0000848	0.0021514	0.0086615

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	7.472e-02	3.118e-02	2.396	0.023011	*
Abslat	-8.887e-05	1.369e-04	-0.649	0.521071	
Spanish	1.308e-02	5.428e-03	2.410	0.022269	*
French	8.935e-03	4.106e-03	2.176	0.037533	*
Brit	6.797e-03	3.648e-03	1.863	0.072235	.
WarDummy	-1.443e-03	2.373e-03	-0.608	0.547746	
LatAmerica	-1.274e-02	5.898e-03	-2.161	0.038827	*
SubSahara	-2.011e-02	6.766e-03	-2.973	0.005775	**
OutwarOr	-4.529e-03	2.064e-03	-2.194	0.036132	*
Area	3.345e-07	7.024e-07	0.476	0.637343	
PrScEnroll	2.019e-02	9.658e-03	2.090	0.045185	*
LifeExp	8.913e-04	2.755e-04	3.236	0.002954	**
GDP60	-1.696e-02	3.230e-03	-5.253	1.14e-05	***
Mining	4.011e-02	1.400e-02	2.864	0.007556	**
EcoOrg	1.114e-03	1.020e-03	1.092	0.283727	
YrsOpen	-2.687e-03	5.356e-03	-0.502	0.619533	
Age	-8.563e-06	2.957e-05	-0.290	0.774109	
Buddha	7.338e-03	6.619e-03	1.109	0.276434	
Catholic	3.075e-03	5.688e-03	0.541	0.592782	
Confucian	7.480e-02	1.510e-02	4.955	2.65e-05	***
EthnoL	1.374e-02	5.140e-03	2.673	0.012047	*
Hindu	-9.667e-02	2.486e-02	-3.889	0.000517	***
Jewish	-7.731e-04	1.057e-02	-0.073	0.942164	
Muslim	1.067e-02	8.639e-03	1.235	0.226576	
PrExports	-5.499e-03	6.736e-03	-0.816	0.420740	
Protestants	-2.801e-03	6.648e-03	-0.421	0.676504	
RuleofLaw	1.161e-02	6.122e-03	1.897	0.067497	.
Popg	-1.143e-01	2.212e-01	-0.517	0.609078	
WorkPop	-5.211e-04	7.502e-03	-0.069	0.945081	
LabForce	3.226e-07	1.036e-07	3.115	0.004023	**
HighEnroll	-1.294e-01	3.839e-02	-3.370	0.002083	**
PublEduPct	1.377e-01	1.172e-01	1.175	0.249311	
RevnCoups	3.207e-03	4.735e-03	0.677	0.503338	
PolRights	1.623e-04	1.954e-03	0.083	0.934362	
CivLib	-2.425e-03	2.406e-03	-1.008	0.321433	
English	-7.070e-03	4.558e-03	-1.551	0.131316	
Foreign	-2.462e-03	3.352e-03	-0.735	0.468281	
RFEXDist	-1.943e-05	3.119e-05	-0.623	0.537900	
EquipInv	1.273e-01	4.451e-02	2.859	0.007652	**
NequipInv	3.669e-02	2.030e-02	1.808	0.080721	.
stdBMP	-1.609e-06	1.172e-05	-0.137	0.891706	
BlMktPm	-6.902e-03	3.740e-03	-1.845	0.074856	.

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.005622 on 30 degrees of freedom
Multiple R-squared: 0.9599, Adjusted R-squared: 0.9052
F-statistic: 17.53 on 41 and 30 DF, p-value: 1.29e-12

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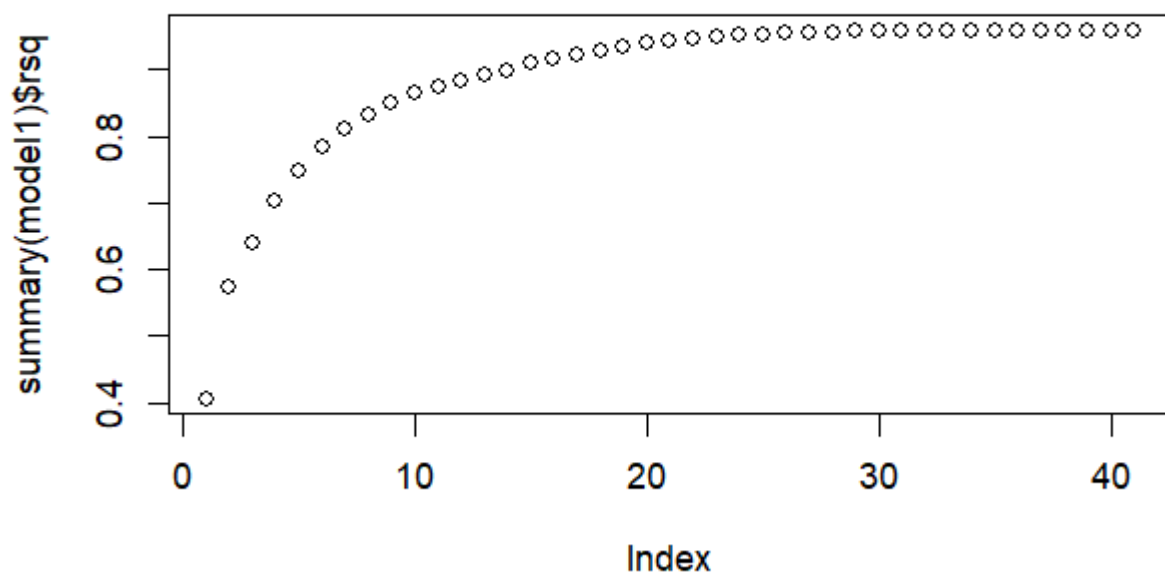
```
# The following is AI generated code to help sort the variables in terms of p-values
summary(model0)$coefficients |>
  as.data.frame() |>
  tibble::rownames_to_column("Variable") |>
  dplyr::arrange(`Pr(>|t|)`)
```

Variable <chr>	Estimate <dbl>	Std. Error <dbl>	t value <dbl>	Pr(> t) <dbl>
GDP60	-1.696439e-02	3.229693e-03	-5.25263126	1.141353e-05
Confucian	7.479622e-02	1.509649e-02	4.95454232	2.648671e-05
Hindu	-9.667198e-02	2.485583e-02	-3.88930864	5.171374e-04
HighEnroll	-1.293723e-01	3.839223e-02	-3.36975121	2.082700e-03
LifeExp	8.913386e-04	2.754818e-04	3.23556248	2.954442e-03
LabForce	3.226487e-07	1.035642e-07	3.11544770	4.022874e-03
SubSahara	-2.011360e-02	6.766495e-03	-2.97252836	5.774878e-03
Mining	4.011120e-02	1.400312e-02	2.86444743	7.556467e-03
Equiplnv	1.272799e-01	4.451343e-02	2.85936039	7.651936e-03
EthnoL	1.373860e-02	5.140217e-03	2.67276608	1.204732e-02
1-10 of 42 rows			Previous 1 2 3 4 5 Next	

We now do model selection by exhaustive search. Note that we have 2^{41} which is approximately 2 trillion possible regressions to run. The leaps package has some smart ways to search over this space by avoiding visiting parts of the space where the optimum cannot exist. It employs a branch and bound algorithm to search more efficiently. This would take a few minutes to run on a laptop. The plot shows the bias-variance tradeoff where the adjusted Rsquared value increases first and then decreases. We can also plot the variables identified using the plot command. Here the model with best adjusted Rsquared value has 27 predictor variables. The plot rendered, though some variable labels are cut off due to the 41-predictor width.

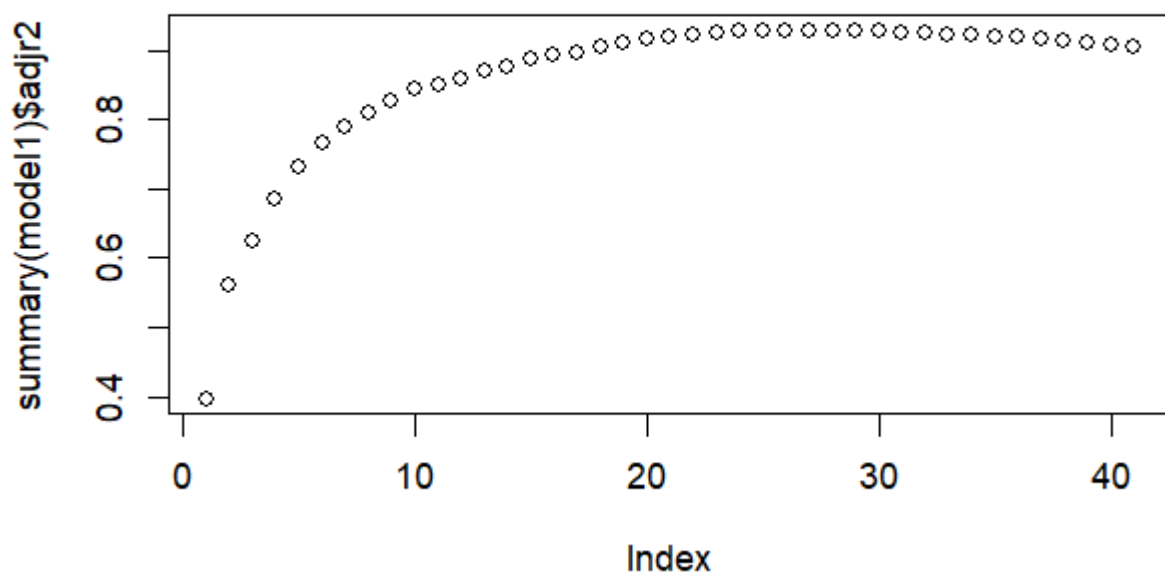
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```
library(leaps)
model1 <- regsubsets(y~.,data=eg1,nvmax=41)
plot(summary(model1)$rsq)
```



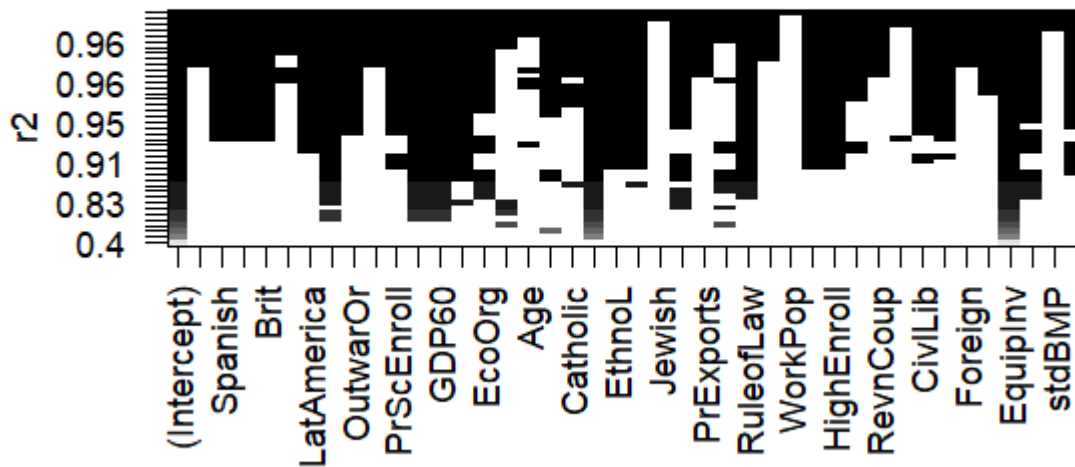
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```
plot(summary(model1)$adjr2)
```



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```
plot(model1, scale=c("r2"))
```



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```
which.max(summary(model1)$adjr2)
```

```
[1] 27
```

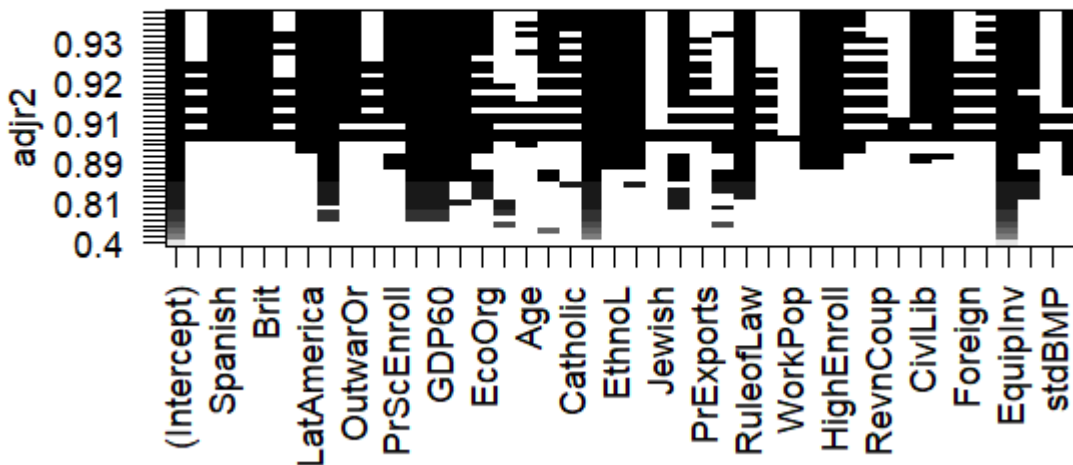
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```
coef(model1,27)
```

(Intercept)	Spanish	French	Brit
6.199916e-02	1.153757e-02	9.805490e-03	6.246981e-03
LatAmerica	SubSahara	OutwarOr	PrScEnroll
-1.097355e-02	-1.692422e-02	-4.280690e-03	1.819797e-02
LifeExp	GDP60	Mining	EcoOrg
9.823127e-04	-1.730846e-02	3.613686e-02	1.007110e-03
Buddha	Catholic	Confucian	EthnoL
9.547808e-03	4.545846e-03	8.293419e-02	1.539070e-02
Hindu	Muslim	RuleofLaw	LabForce
-1.010234e-01	1.085449e-02	1.029866e-02	3.672291e-07
HighEnroll	PublEduPct	CivLib	English
-1.234089e-01	1.043950e-01	-2.430588e-03	-5.836167e-03
RFEXDist	EquipInv	NequipInv	BlMktPm
-2.274696e-05	1.378925e-01	4.206545e-02	-6.217038e-03

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```
plot(model1,scale=c("adjr2"))
```



I prompted the Posit Assistant to make a plot with all variables shown on the axis “Can you plot with all 41 variables shown clearly?”. It gave the code below.

Hide

```
library(ggplot2)
library(tidyr)
library(dplyr)

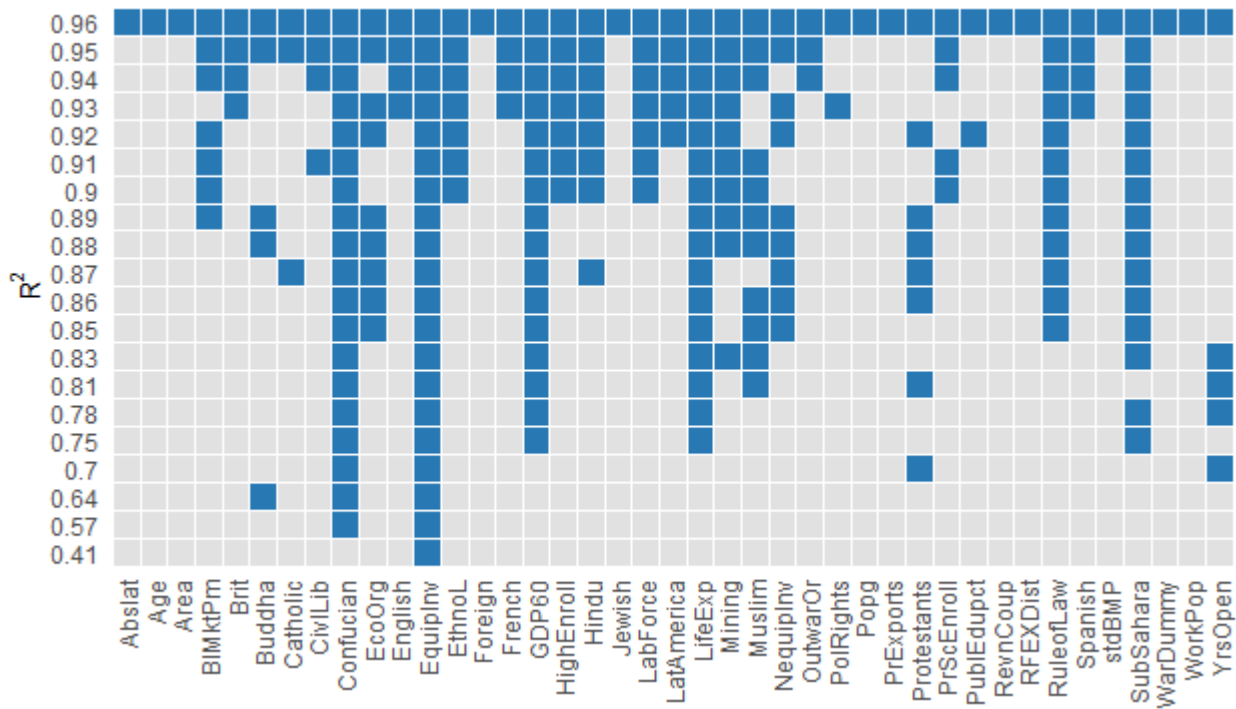
s1 <- summary(model1)

# Build a long data frame: rows = subset size, cols = variable, value = included (T/F)
which_df <- as.data.frame(s1$which) |>
  mutate(r2 = round(s1$rsq, 3), subset_size = row_number()) |>
  pivot_longer(-c(r2, subset_size), names_to = "Variable", values_to = "Included") |>
  filter(Variable != "(Intercept)")

options(repr.plot.width = 16, repr.plot.height = 10)

ggplot(which_df, aes(x = Variable, y = factor(round(r2, 2)), fill = Included)) +
  geom_tile(color = "white", linewidth = 0.3) +
  scale_fill_manual(values = c("TRUE" = "#2c7bb6", "FALSE" = "grey90")) +
  theme_minimal(base_size = 9) +
  theme(
    axis.text.x = element_text(angle = 90, hjust = 1, vjust = 0.5, size = 8),
    axis.text.y = element_text(size = 8),
    legend.position = "none",
    panel.grid = element_blank(),
    plot.title = element_text(size = 10, face = "bold")
  ) +
  labs(x = NULL, y = expression(R^2), title = "Best Subsets by R2 (Exhaustive Search)")
```

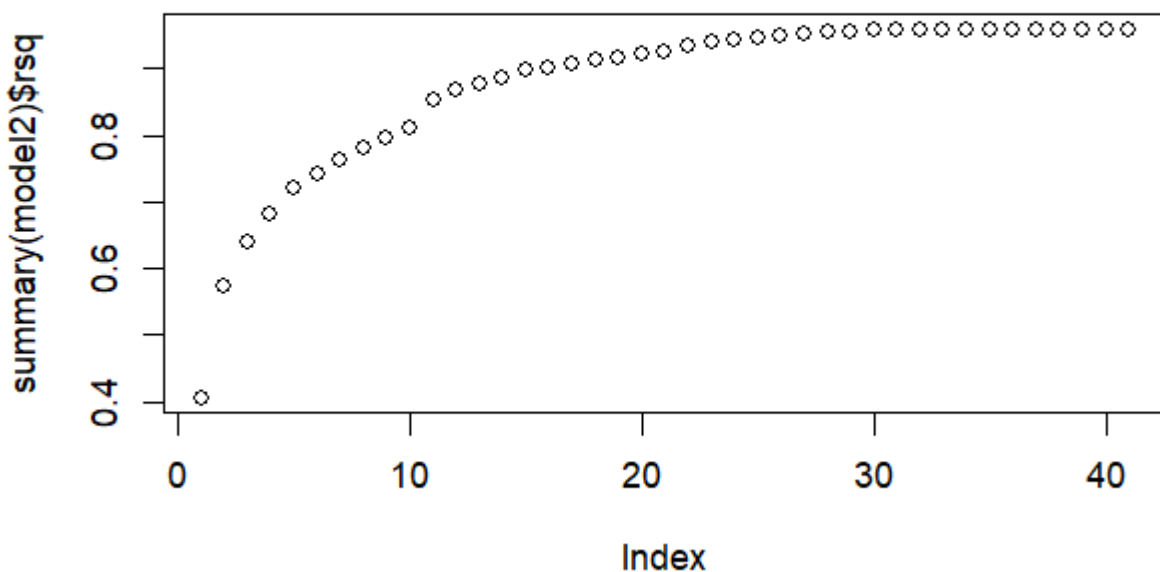

Best Subsets by R² (Exhaustive Search)



We next use the forward stepwise selection method which runs much faster as should be expected. Note that the results are not identical to what we obtained with the exhaustive selection approach. Here the model with the best adjusted Rsquared has 29 predictor variables.

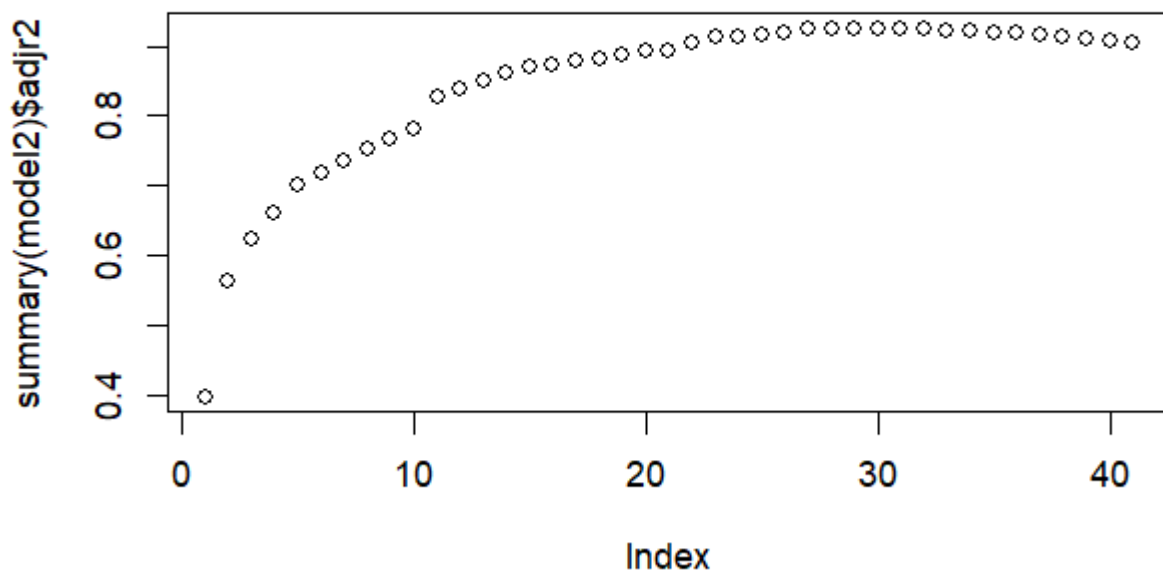
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```
model2 <- regsubsets(y~.,data=eg1,nvmax=41,method="forward")
plot(summary(model2)$rsq)
```



Hide

```
plot(summary(model2)$adjr2)
```



Hide

```
which.max(summary(model12)$adjr2)
```

```
[1] 29
```

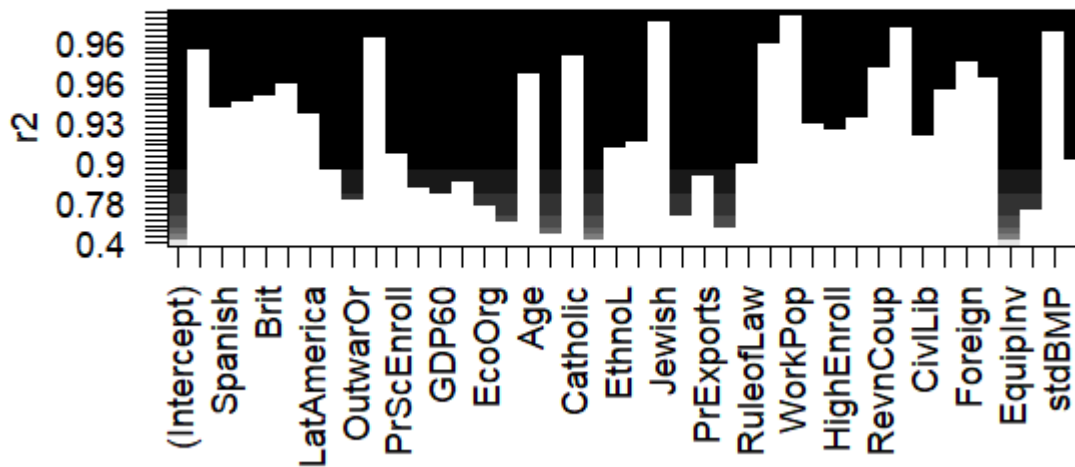
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```
coef(model12,29)
```

```
(Intercept)      Spanish      French      Brit
7.642126e-02  1.130845e-02  7.942888e-03  6.022931e-03
WarDummy      LatAmerica      SubSahara      OutwarOr
-1.850683e-03 -1.127019e-02 -2.079125e-02 -4.663622e-03
PrScEnroll      LifeExp      GDP60      Mining
2.031293e-02  8.184873e-04 -1.745551e-02  3.638828e-02
EcoOrg      YrsOpen      Buddha      Confucian
1.083773e-03 -1.377063e-03  7.279949e-03  7.178168e-02
EthnoL      Hindu      Muslim      PrExports
1.511355e-02 -1.000899e-01  6.584084e-03 -3.152428e-03
Protestants      RuleofLaw      LabForce      HighEnroll
-5.100076e-03  1.153917e-02  3.339390e-07 -1.134317e-01
PublEduPct      CivlLib      English      EquipInv
1.189474e-01 -2.342815e-03 -4.982871e-03  1.207018e-01
NequipInv      BLMktPm
3.634847e-02 -6.105242e-03
```

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```
plot(model12,scale=c("r2"))
```



The results indicate that with model 1, we have 1) Equiplnv, 2) Confucian, Equiplnv, 3) Buddha, Confucian, Equiplnv, 4) YrsOpen, Confucian, Protestants, Equiplnv while for model 2, we have 1) Equiplnv, 2) Confucian, Equiplnv, 3) Buddha, Confucian, Equiplnv, 4) Buddha, Protestants, Equiplnv, Confucian and so on. The results are different from the two models. Also observe that the results are different to choosing variables simply based on smallest p-values to highest p-values.

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```
summary(model1)$which
```

	(Intercept)	Abslat	Spanish	French	Brit	WarDummy	LatAmerica	
1	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	
2	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	
3	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	
4	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	
5	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	
6	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	
7	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	
8	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	
9	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	
10	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	
11	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	
12	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	
13	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	
14	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	
15	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	
16	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	
17	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE	
18	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE	
19	TRUE	FALSE	TRUE	TRUE	TRUE	FALSE	TRUE	
20	TRUE	FALSE	TRUE	TRUE	TRUE	FALSE	TRUE	
21	TRUE	FALSE	TRUE	TRUE	TRUE	FALSE	TRUE	
22	TRUE	FALSE	TRUE	TRUE	TRUE	FALSE	TRUE	
23	TRUE	FALSE	TRUE	TRUE	TRUE	FALSE	TRUE	
	SubSahara	OutwarOr	Area	PrScEnroll	LifeExp	GDP60	Mining	EcoOrg
1	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
2	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
3	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
4	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
5	TRUE	FALSE	FALSE	FALSE	TRUE	TRUE	FALSE	FALSE
6	TRUE	FALSE	FALSE	FALSE	TRUE	TRUE	FALSE	FALSE
7	FALSE	FALSE	FALSE	FALSE	TRUE	TRUE	FALSE	FALSE
8	TRUE	FALSE	FALSE	FALSE	TRUE	TRUE	TRUE	FALSE
9	TRUE	FALSE	FALSE	FALSE	TRUE	TRUE	FALSE	TRUE
10	TRUE	FALSE	FALSE	FALSE	TRUE	TRUE	FALSE	TRUE
11	TRUE	FALSE	FALSE	FALSE	TRUE	TRUE	FALSE	TRUE
12	TRUE	FALSE	FALSE	FALSE	TRUE	TRUE	TRUE	TRUE
13	TRUE	FALSE	FALSE	FALSE	TRUE	TRUE	TRUE	TRUE
14	TRUE	FALSE	FALSE	TRUE	TRUE	TRUE	TRUE	FALSE
15	TRUE	FALSE	FALSE	TRUE	TRUE	TRUE	TRUE	FALSE
16	TRUE	FALSE	FALSE	TRUE	TRUE	TRUE	TRUE	FALSE
17	TRUE	FALSE	FALSE	FALSE	TRUE	TRUE	TRUE	TRUE
18	TRUE	FALSE	FALSE	FALSE	TRUE	TRUE	TRUE	TRUE
19	TRUE	FALSE	FALSE	FALSE	TRUE	TRUE	TRUE	TRUE
20	TRUE	TRUE	FALSE	TRUE	TRUE	TRUE	TRUE	FALSE
21	TRUE	TRUE	FALSE	TRUE	TRUE	TRUE	TRUE	FALSE
22	TRUE	TRUE	FALSE	TRUE	TRUE	TRUE	TRUE	FALSE
23	TRUE	TRUE	FALSE	TRUE	TRUE	TRUE	TRUE	FALSE
	YrsOpen	Age	Buddha	Catholic	Confucian	EthnoL	Hindu	Jewish
1	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
2	FALSE	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE
3	FALSE	FALSE	TRUE	FALSE	TRUE	FALSE	FALSE	FALSE
4	TRUE	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE
5	FALSE	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE
6	TRUE	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE

7	TRUE	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE
8	TRUE	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE
9	FALSE	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE
10	FALSE	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE
11	FALSE	FALSE	FALSE	TRUE	TRUE	FALSE	TRUE	FALSE
12	FALSE	FALSE	TRUE	FALSE	TRUE	FALSE	FALSE	FALSE
13	FALSE	FALSE	TRUE	FALSE	TRUE	FALSE	FALSE	FALSE
14	FALSE	FALSE	FALSE	FALSE	TRUE	TRUE	TRUE	FALSE
15	FALSE	FALSE	FALSE	FALSE	TRUE	TRUE	TRUE	FALSE
16	FALSE	FALSE	FALSE	FALSE	TRUE	TRUE	TRUE	FALSE
17	FALSE	FALSE	FALSE	FALSE	TRUE	TRUE	TRUE	FALSE
18	FALSE	TRUE	FALSE	FALSE	TRUE	TRUE	TRUE	FALSE
19	FALSE	FALSE	FALSE	FALSE	TRUE	TRUE	TRUE	FALSE
20	FALSE	FALSE	FALSE	FALSE	TRUE	TRUE	TRUE	FALSE
21	FALSE	FALSE	FALSE	FALSE	TRUE	TRUE	TRUE	FALSE
22	FALSE	FALSE	FALSE	FALSE	TRUE	TRUE	TRUE	FALSE
23	FALSE	FALSE	TRUE	FALSE	TRUE	TRUE	TRUE	FALSE
	Muslim	PrExports	Protestants	RuleofLaw	Popg	WorkPop	LabForce	
1	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
2	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
3	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
4	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE
5	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
6	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
7	TRUE	FALSE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE
8	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
9	TRUE	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE	FALSE
10	TRUE	FALSE	TRUE	TRUE	FALSE	FALSE	FALSE	FALSE
11	FALSE	FALSE	TRUE	TRUE	FALSE	FALSE	FALSE	FALSE
12	TRUE	FALSE	TRUE	TRUE	FALSE	FALSE	FALSE	FALSE
13	TRUE	FALSE	TRUE	TRUE	FALSE	FALSE	FALSE	FALSE
14	TRUE	FALSE	FALSE	TRUE	FALSE	FALSE	TRUE	
15	TRUE	FALSE	FALSE	TRUE	FALSE	FALSE	TRUE	
16	TRUE	FALSE	FALSE	TRUE	FALSE	FALSE	TRUE	
17	FALSE	FALSE	TRUE	TRUE	FALSE	FALSE	TRUE	
18	FALSE	FALSE	TRUE	TRUE	FALSE	FALSE	TRUE	
19	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE	TRUE	
20	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE	TRUE	
21	TRUE	FALSE	FALSE	TRUE	FALSE	FALSE	TRUE	
22	TRUE	FALSE	FALSE	TRUE	FALSE	FALSE	TRUE	
23	TRUE	FALSE	FALSE	TRUE	FALSE	FALSE	TRUE	
	HighEnroll	PublEduPct	RevnCoup	PolRights	CivlLib	English		
1	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	
2	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	
3	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	
4	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	
5	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	
6	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	
7	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	
8	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	
9	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	
10	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	
11	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	
12	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	
13	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	
14	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	

15	TRUE	FALSE	FALSE	FALSE	TRUE	FALSE
16	TRUE	FALSE	FALSE	FALSE	TRUE	TRUE
17	TRUE	TRUE	FALSE	FALSE	FALSE	FALSE
18	TRUE	TRUE	FALSE	FALSE	FALSE	FALSE
19	TRUE	FALSE	FALSE	TRUE	FALSE	TRUE
20	TRUE	FALSE	FALSE	FALSE	TRUE	TRUE
21	TRUE	FALSE	FALSE	FALSE	TRUE	TRUE
22	TRUE	FALSE	FALSE	FALSE	TRUE	TRUE
23	TRUE	FALSE	FALSE	FALSE	TRUE	TRUE

	Foreign	RFEXDist	EquipInv	NequipInv	stdBMP	BlMktPm
--	---------	----------	----------	-----------	--------	---------

1	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE
2	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE
3	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE
4	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE
5	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE
6	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE
7	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE
8	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE
9	FALSE	FALSE	TRUE	TRUE	FALSE	FALSE
10	FALSE	FALSE	TRUE	TRUE	FALSE	FALSE
11	FALSE	FALSE	TRUE	TRUE	FALSE	FALSE
12	FALSE	FALSE	TRUE	TRUE	FALSE	FALSE
13	FALSE	FALSE	TRUE	TRUE	FALSE	TRUE
14	FALSE	FALSE	TRUE	FALSE	FALSE	TRUE
15	FALSE	FALSE	TRUE	FALSE	FALSE	TRUE
16	FALSE	FALSE	TRUE	FALSE	FALSE	TRUE
17	FALSE	FALSE	TRUE	TRUE	FALSE	TRUE
18	FALSE	FALSE	TRUE	TRUE	FALSE	TRUE
19	FALSE	FALSE	TRUE	TRUE	FALSE	FALSE
20	FALSE	FALSE	TRUE	TRUE	FALSE	FALSE
21	FALSE	FALSE	TRUE	FALSE	FALSE	TRUE
22	FALSE	FALSE	TRUE	TRUE	FALSE	TRUE
23	FALSE	FALSE	TRUE	TRUE	FALSE	TRUE

[reached 'max' / getOption("max.print") -- omitted 18 rows]

Hide

```
summary(model2)$which
```

	(Intercept)	Abslat	Spanish	French	Brit	WarDummy	LatAmerica	
1	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	
2	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	
3	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	
4	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	
5	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	
6	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	
7	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	
8	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	
9	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	
10	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	
11	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	
12	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	
13	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	
14	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	
15	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	
16	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	
17	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	
18	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	
19	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	
20	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	
21	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	
22	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	
23	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	
	SubSahara	OutwarOr	Area	PrScEnroll	LifeExp	GDP60	Mining	EcoOrg
1	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
2	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
3	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
4	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
5	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
6	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
7	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
8	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE
9	FALSE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE
10	FALSE	TRUE	FALSE	FALSE	FALSE	TRUE	FALSE	TRUE
11	FALSE	TRUE	FALSE	FALSE	TRUE	TRUE	FALSE	TRUE
12	FALSE	TRUE	FALSE	FALSE	TRUE	TRUE	TRUE	TRUE
13	FALSE	TRUE	FALSE	FALSE	TRUE	TRUE	TRUE	TRUE
14	TRUE	TRUE	FALSE	FALSE	TRUE	TRUE	TRUE	TRUE
15	TRUE	TRUE	FALSE	FALSE	TRUE	TRUE	TRUE	TRUE
16	TRUE	TRUE	FALSE	FALSE	TRUE	TRUE	TRUE	TRUE
17	TRUE	TRUE	FALSE	TRUE	TRUE	TRUE	TRUE	TRUE
18	TRUE	TRUE	FALSE	TRUE	TRUE	TRUE	TRUE	TRUE
19	TRUE	TRUE	FALSE	TRUE	TRUE	TRUE	TRUE	TRUE
20	TRUE	TRUE	FALSE	TRUE	TRUE	TRUE	TRUE	TRUE
21	TRUE	TRUE	FALSE	TRUE	TRUE	TRUE	TRUE	TRUE
22	TRUE	TRUE	FALSE	TRUE	TRUE	TRUE	TRUE	TRUE
23	TRUE	TRUE	FALSE	TRUE	TRUE	TRUE	TRUE	TRUE
	YrsOpen	Age	Buddha	Catholic	Confucian	EthnoL	Hindu	Jewish
1	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
2	FALSE	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE
3	FALSE	FALSE	TRUE	FALSE	TRUE	FALSE	FALSE	FALSE
4	FALSE	FALSE	TRUE	FALSE	TRUE	FALSE	FALSE	FALSE
5	TRUE	FALSE	TRUE	FALSE	TRUE	FALSE	FALSE	FALSE
6	TRUE	FALSE	TRUE	FALSE	TRUE	FALSE	FALSE	FALSE

[illegible]

15	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
16	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
17	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
18	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
19	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
20	FALSE	FALSE	FALSE	FALSE	TRUE	FALSE
21	TRUE	FALSE	FALSE	FALSE	TRUE	FALSE
22	TRUE	FALSE	FALSE	FALSE	TRUE	FALSE
23	TRUE	TRUE	FALSE	FALSE	TRUE	FALSE

	Foreign	RFEXDist	EquipInv	NequipInv	stdBMP	BlMktPm
--	---------	----------	----------	-----------	--------	---------

1	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE
2	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE
3	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE
4	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE
5	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE
6	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE
7	FALSE	FALSE	TRUE	TRUE	FALSE	FALSE
8	FALSE	FALSE	TRUE	TRUE	FALSE	FALSE
9	FALSE	FALSE	TRUE	TRUE	FALSE	FALSE
10	FALSE	FALSE	TRUE	TRUE	FALSE	FALSE
11	FALSE	FALSE	TRUE	TRUE	FALSE	FALSE
12	FALSE	FALSE	TRUE	TRUE	FALSE	FALSE
13	FALSE	FALSE	TRUE	TRUE	FALSE	FALSE
14	FALSE	FALSE	TRUE	TRUE	FALSE	FALSE
15	FALSE	FALSE	TRUE	TRUE	FALSE	FALSE
16	FALSE	FALSE	TRUE	TRUE	FALSE	TRUE
17	FALSE	FALSE	TRUE	TRUE	FALSE	TRUE
18	FALSE	FALSE	TRUE	TRUE	FALSE	TRUE
19	FALSE	FALSE	TRUE	TRUE	FALSE	TRUE
20	FALSE	FALSE	TRUE	TRUE	FALSE	TRUE
21	FALSE	FALSE	TRUE	TRUE	FALSE	TRUE
22	FALSE	FALSE	TRUE	TRUE	FALSE	TRUE
23	FALSE	FALSE	TRUE	TRUE	FALSE	TRUE

```
[ reached 'max' /getOption("max.print") -- omitted 18 rows ]
```

LASSO model: The results indicate that for variables such as EquipInv, YrsOpen, Confucian show up for many values of lambda while some other variables such as CivilLib, Abslat, Popg do not show up as often. Such results help illustrate the reliability of predictors for economic growth and also help cast doubts on the robustness of the results for certain variables which might be proposed as being correlated with growth but do not show up often. You can also look at the signs of the beta coefficients to see how they relate to growth.

Hide

```
library(glmnet)
x <- eg1[,c(2:42)] # here you do not need to use as.matrix(eg1[,c(2:42)]) as the variables are
# stored as numeric values; code will still work correctly if you use as.matrix(eg1[,c(2:42)])
model3 <- glmnet(x, eg1$y)
model3
```

Call: glmnet(x = x, y = eg1\$y)

	Df	%Dev	Lambda
1	0	0.00	0.0115300
2	1	6.87	0.0105100
3	2	13.30	0.0095730
4	3	19.31	0.0087230
5	3	26.69	0.0079480
6	3	32.82	0.0072420
7	3	37.90	0.0065990
8	4	42.67	0.0060120
9	4	46.81	0.0054780
10	4	50.25	0.0049920
11	4	53.10	0.0045480
12	5	55.67	0.0041440
13	5	57.92	0.0037760
14	7	60.01	0.0034400
15	7	62.46	0.0031350
16	9	64.55	0.0028560
17	10	66.56	0.0026030
18	10	68.33	0.0023710
19	11	69.86	0.0021610
20	12	71.49	0.0019690
21	13	73.00	0.0017940
22	13	74.28	0.0016350
23	14	75.46	0.0014890
24	15	76.58	0.0013570
25	17	77.91	0.0012360
26	19	79.34	0.0011270
27	20	81.10	0.0010270
28	20	82.63	0.0009353
29	21	83.90	0.0008522
30	23	85.01	0.0007765
31	24	85.99	0.0007075
32	24	86.86	0.0006447
33	26	87.64	0.0005874
34	26	88.39	0.0005352
35	26	89.00	0.0004877
36	26	89.50	0.0004444
37	27	89.94	0.0004049
38	28	90.34	0.0003689
39	29	90.80	0.0003361
40	30	91.22	0.0003063
41	30	91.79	0.0002791
42	30	92.28	0.0002543
43	30	92.68	0.0002317
44	32	93.02	0.0002111
45	32	93.36	0.0001924
46	34	93.64	0.0001753
47	35	93.91	0.0001597
48	35	94.22	0.0001455
49	35	94.48	0.0001326
50	34	94.71	0.0001208
51	35	94.89	0.0001101

```

52 35 95.05 0.0001003
53 35 95.19 0.0000914
54 37 95.30 0.0000833
55 37 95.40 0.0000759
56 36 95.48 0.0000691
57 36 95.54 0.0000630
58 36 95.60 0.0000574
59 36 95.64 0.0000523
60 36 95.68 0.0000476
61 36 95.71 0.0000434
62 36 95.74 0.0000396
63 36 95.76 0.0000360
64 38 95.78 0.0000328
65 40 95.81 0.0000299
66 41 95.84 0.0000273
67 41 95.86 0.0000248
68 41 95.88 0.0000226
69 41 95.90 0.0000206
70 41 95.92 0.0000188
71 41 95.93 0.0000171
72 41 95.94 0.0000156
73 41 95.95 0.0000142
74 41 95.95 0.0000130
75 41 95.96 0.0000118
76 41 95.96 0.0000108
77 41 95.97 0.0000098
78 40 95.97 0.0000089
79 40 95.97 0.0000081
80 40 95.98 0.0000074
81 40 95.98 0.0000068
82 40 95.98 0.0000062
83 40 95.98 0.0000056
84 40 95.98 0.0000051
85 40 95.98 0.0000047
86 40 95.99 0.0000042

```

Hide

```
summary(model3)
```

	Length	Class	Mode
a0	86	-none-	numeric
beta	3526	dgCMatrix	S4
df	86	-none-	numeric
dim	2	-none-	numeric
lambda	86	-none-	numeric
dev.ratio	86	-none-	numeric
nulldev	1	-none-	numeric
npasses	1	-none-	numeric
jerr	1	-none-	numeric
offset	1	-none-	logical
call	3	-none-	call
nobs	1	-none-	numeric

Hide

```
model3$df
```

```
[1] 0 1 2 3 3 3 3 4 4 4 4 5 5 7 7 9 10 10 11 12 13
[22] 13 14 15 17 19 20 20 21 23 24 24 26 26 26 26 27 28 29 30 30 30
[43] 30 32 32 34 35 35 35 34 35 35 35 37 37 36 36 36 36 36 36 36
[64] 38 40 41 41 41 41 41 41 41 41 41 41 41 40 40 40 40 40 40 40
[85] 40 40
```

[Hide](#)

```
model3$beta["EquipInv",]
```

s0	s1	s2	s3	s4	s5
0.00000000	0.02989896	0.05202483	0.06760673	0.08042867	0.09212352
s6	s7	s8	s9	s10	s11
0.10277946	0.11258867	0.12154191	0.12971174	0.13715581	0.14406270
s12	s13	s14	s15	s16	s17
0.15040049	0.15383321	0.15867593	0.16258346	0.16444915	0.16555791
s18	s19	s20	s21	s22	s23
0.16661977	0.16734255	0.16832148	0.16948025	0.17126770	0.17293008
s24	s25	s26	s27	s28	s29
0.17351524	0.17220683	0.16702669	0.16130824	0.15651196	0.15187566
s30	s31	s32	s33	s34	s35
0.14705527	0.14234527	0.13738774	0.13193100	0.12706968	0.12274067
s36	s37	s38	s39	s40	s41
0.11929709	0.11510558	0.11298039	0.11149575	0.11069796	0.10979416
s42	s43	s44	s45	s46	s47
0.10902316	0.10822285	0.10826583	0.10805193	0.10978025	0.11122019
s48	s49	s50	s51	s52	s53
0.11265440	0.11364832	0.11454541	0.11517577	0.11571790	0.11637044
s54	s55	s56	s57	s58	s59
0.11769020	0.11837771	0.11911269	0.11981383	0.12046601	0.12107996
s60	s61	s62	s63	s64	s65
0.12163600	0.12214121	0.12259898	0.12310262	0.12363149	0.12418720
s66	s67	s68	s69	s70	s71
0.12446285	0.12470910	0.12491215	0.12509626	0.12525557	0.12540204
s72	s73	s74	s75	s76	s77
0.12553559	0.12566579	0.12577917	0.12588798	0.12597718	0.12607745
s78	s79	s80	s81	s82	s83
0.12615294	0.12619169	0.12622857	0.12626553	0.12629465	0.12632607
s84	s85				
0.12635177	0.12637656				

[Hide](#)

```
model3$beta["YrsOpen",]
```

s0	s1	s2	s3
0.0000000000	0.0000000000	0.0008411755	0.0023489833
s4	s5	s6	s7
0.0034594779	0.0044693868	0.0053895748	0.0060181961
s8	s9	s10	s11
0.0065196996	0.0069746659	0.0073892098	0.0075017961
s12	s13	s14	s15
0.0074597160	0.0073590302	0.0075806824	0.0077767586
s16	s17	s18	s19
0.0077803959	0.0076243929	0.0075313623	0.0075309642
s20	s21	s22	s23
0.0074339216	0.0073339037	0.0072385791	0.0071058236
s24	s25	s26	s27
0.0071862421	0.0069866982	0.0069500973	0.0070605306
s28	s29	s30	s31
0.0071729800	0.0072834395	0.0073558363	0.0071813060
s32	s33	s34	s35
0.0070002310	0.0065755290	0.0062229300	0.0058955689
s36	s37	s38	s39
0.0056337588	0.0053449853	0.0048998590	0.0043785966
s40	s41	s42	s43
0.0039529149	0.0035579601	0.0032070905	0.0028786714
s44	s45	s46	s47
0.0024366683	0.0019851088	0.0016789639	0.0014733169
s48	s49	s50	s51
0.0012866425	0.0010697702	0.0008549662	0.0006260712
s52	s53	s54	s55
0.0004303385	0.0002377157	0.0000396649	0.0000000000
s56	s57	s58	s59
0.0000000000	0.0000000000	0.0000000000	0.0000000000
s60	s61	s62	s63
0.0000000000	0.0000000000	0.0000000000	0.0000000000
s64	s65	s66	s67
-0.0000347153	-0.0002031494	-0.0004074152	-0.0005981505
s68	s69	s70	s71
-0.0007729291	-0.0009349521	-0.0010811568	-0.0012163362
s72	s73	s74	s75
-0.0013327691	-0.0014456482	-0.0015488999	-0.0016386196
s76	s77	s78	s79
-0.0017214955	-0.0018029329	-0.0018718677	-0.0019346070
s80	s81	s82	s83
-0.0019868902	-0.0020492098	-0.0020877233	-0.0021380883
s84	s85		
-0.0021754170	-0.0022126840		

[Hide](#)

```
model3$beta["Confucian",]
```

s0	s1	s2	s3	s4
0.0000000000	0.0000000000	0.0000000000	0.0001621431	0.0073470601
s5	s6	s7	s8	s9
0.0138931647	0.0198577305	0.0247184314	0.0289443053	0.0327943465
s10	s11	s12	s13	s14
0.0363023598	0.0394825702	0.0423720756	0.0451703608	0.0473677815
s15	s16	s17	s18	s19
0.0494006357	0.0513695192	0.0532800898	0.0550396318	0.0568168456
s20	s21	s22	s23	s24
0.0580045783	0.0590249207	0.0597445690	0.0602435836	0.0602429141
s25	s26	s27	s28	s29
0.0596761196	0.0592215257	0.0587600550	0.0583418572	0.0577364259
s30	s31	s32	s33	s34
0.0568774527	0.0562989043	0.0556491892	0.0554179214	0.0552196992
s35	s36	s37	s38	s39
0.0550505400	0.0547964120	0.0546579435	0.0549502195	0.0555138745
s40	s41	s42	s43	s44
0.0567310639	0.0577869720	0.0587342490	0.0595534109	0.0601290032
s45	s46	s47	s48	s49
0.0605148323	0.0613936418	0.0624443098	0.0635110808	0.0642446690
s50	s51	s52	s53	s54
0.0649079908	0.0654469803	0.0659337479	0.0663722431	0.0666905089
s55	s56	s57	s58	s59
0.0669304933	0.0671519439	0.0673599910	0.0675530982	0.0677332176
s60	s61	s62	s63	s64
0.0678979832	0.0680485327	0.0681856232	0.0684322068	0.0688534260
s65	s66	s67	s68	s69
0.0693838201	0.0698945900	0.0703152379	0.0706821886	0.0710215912
s70	s71	s72	s73	s74
0.0713174773	0.0715927905	0.0718321509	0.0720705211	0.0722813913
s75	s76	s77	s78	s79
0.0724716965	0.0726395166	0.0728104598	0.0729514088	0.0730629130
s80	s81	s82	s83	s84
0.0731568200	0.0732688430	0.0733399001	0.0734313972	0.0735005287
s85				
0.0735691984				

[Hide](#)

```
model3$beta["CivlLib",]
```

s0	s1	s2	s3
0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00
s4	s5	s6	s7
0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00
s8	s9	s10	s11
0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00
s12	s13	s14	s15
0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00
s16	s17	s18	s19
0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00
s20	s21	s22	s23
0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00
s24	s25	s26	s27
0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00
s28	s29	s30	s31
0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00
s32	s33	s34	s35
0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00
s36	s37	s38	s39
0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00
s40	s41	s42	s43
0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00
s44	s45	s46	s47
0.000000e+00	-1.329499e-05	-1.584215e-04	-3.445745e-04
s48	s49	s50	s51
-5.151464e-04	-6.644557e-04	-8.028445e-04	-9.229976e-04
s52	s53	s54	s55
-1.020101e-03	-1.110300e-03	-1.196981e-03	-1.264754e-03
s56	s57	s58	s59
-1.327613e-03	-1.388443e-03	-1.446220e-03	-1.502062e-03
s60	s61	s62	s63
-1.552938e-03	-1.599233e-03	-1.641196e-03	-1.681093e-03
s64	s65	s66	s67
-1.739457e-03	-1.793881e-03	-1.845063e-03	-1.891035e-03
s68	s69	s70	s71
-1.932224e-03	-1.970819e-03	-2.004646e-03	-2.036117e-03
s72	s73	s74	s75
-2.062775e-03	-2.089466e-03	-2.113558e-03	-2.134795e-03
s76	s77	s78	s79
-2.153822e-03	-2.168417e-03	-2.175046e-03	-2.180725e-03
s80	s81	s82	s83
-2.185444e-03	-2.190700e-03	-2.194135e-03	-2.198425e-03
s84	s85		
-2.201629e-03	-2.204813e-03		

[Hide](#)

```
model3$beta["Abslat",]
```

s0	s1	s2	s3
0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00
s4	s5	s6	s7
0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00
s8	s9	s10	s11
0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00
s12	s13	s14	s15
0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00
s16	s17	s18	s19
0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00
s20	s21	s22	s23
0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00
s24	s25	s26	s27
0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00
s28	s29	s30	s31
0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00
s32	s33	s34	s35
0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00
s36	s37	s38	s39
0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00
s40	s41	s42	s43
0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00
s44	s45	s46	s47
0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00
s48	s49	s50	s51
0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00
s52	s53	s54	s55
0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00
s56	s57	s58	s59
0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00
s60	s61	s62	s63
0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00
s64	s65	s66	s67
-1.622437e-06	-7.518399e-06	-1.431802e-05	-2.042576e-05
s68	s69	s70	s71
-2.591297e-05	-3.102535e-05	-3.556897e-05	-3.979984e-05
s72	s73	s74	s75
-4.337126e-05	-4.690859e-05	-5.016168e-05	-5.296114e-05
s76	s77	s78	s79
-5.555386e-05	-5.806661e-05	-6.007978e-05	-6.195865e-05
s80	s81	s82	s83
-6.347919e-05	-6.536241e-05	-6.648517e-05	-6.799466e-05
s84	s85		
-6.910986e-05	-7.022663e-05		

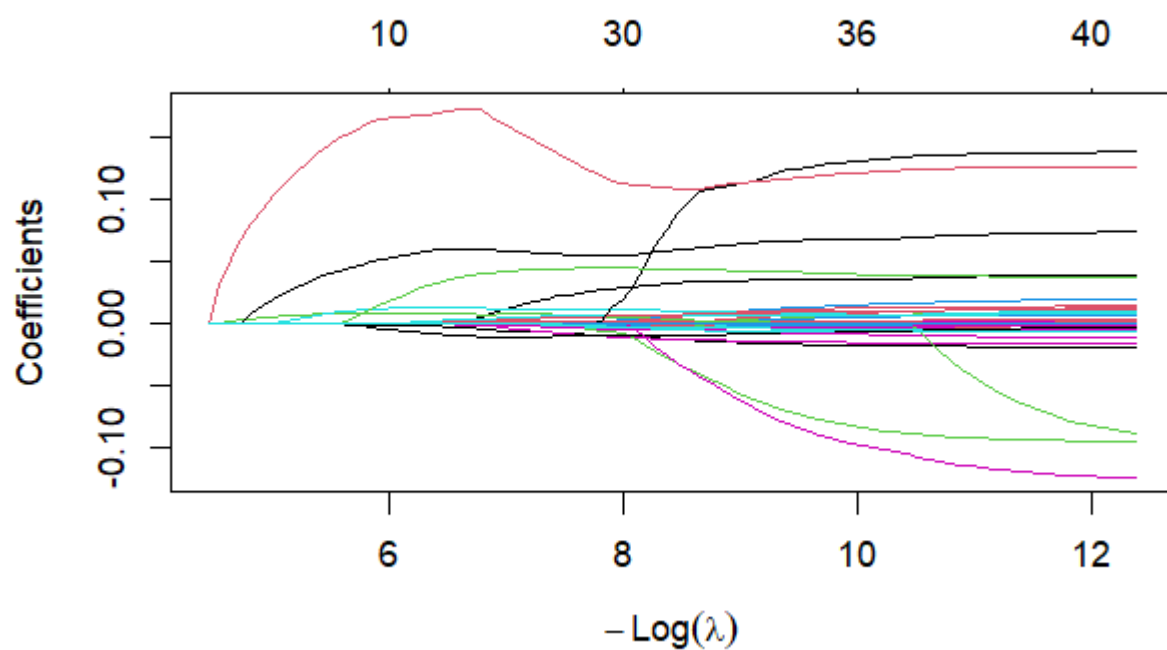
[Hide](#)

```
model3$beta["Popg",]
```


s0	s1	s2	s3	s4
0.000000000	0.000000000	0.000000000	0.000000000	0.000000000
s5	s6	s7	s8	s9
0.000000000	0.000000000	0.000000000	0.000000000	0.000000000
s10	s11	s12	s13	s14
0.000000000	0.000000000	0.000000000	0.000000000	0.000000000
s15	s16	s17	s18	s19
0.000000000	0.000000000	0.000000000	0.000000000	0.000000000
s20	s21	s22	s23	s24
0.000000000	0.000000000	0.000000000	0.000000000	0.000000000
s25	s26	s27	s28	s29
0.000000000	0.000000000	0.000000000	0.000000000	0.000000000
s30	s31	s32	s33	s34
0.000000000	0.000000000	0.000000000	0.000000000	0.000000000
s35	s36	s37	s38	s39
0.000000000	0.000000000	0.000000000	0.000000000	0.000000000
s40	s41	s42	s43	s44
0.000000000	0.000000000	0.000000000	0.000000000	0.000000000
s45	s46	s47	s48	s49
0.000000000	0.000000000	0.000000000	0.000000000	0.000000000
s50	s51	s52	s53	s54
0.000000000	0.000000000	0.000000000	0.000000000	0.000000000
s55	s56	s57	s58	s59
0.000000000	0.000000000	0.000000000	0.000000000	0.000000000
s60	s61	s62	s63	s64
0.000000000	0.000000000	0.000000000	0.000000000	0.000000000
s65	s66	s67	s68	s69
-0.004680145	-0.014078839	-0.022353824	-0.029778758	-0.036699331
s70	s71	s72	s73	s74
-0.042813656	-0.048513898	-0.053399988	-0.058267837	-0.062643931
s75	s76	s77	s78	s79
-0.066499726	-0.069974950	-0.073461972	-0.076305220	-0.078738534
s80	s81	s82	s83	s84
-0.080740069	-0.083200598	-0.084677952	-0.086657997	-0.088119362
s85				
-0.089582002				

[Hide](#)

```
plot(model3,xvar="lambda")
```



In the display below . denotes 0 and | denotes nonzero. For example, growth rate seems to be positively related to school enrollment rates and negatively related to the initial 1960 level of real per capita GDP.

Hide

model3\$beta

41 x 86 sparse Matrix of class "dgCMatrix"

Abslat	.	.	.	.	.
Spanish	.	.	.	.	.
French	.	.	.	.	.
Brit	.	.	.	.	.
WarDummy	.	.	.	.	.
LatAmerica	.	.	.	.	.
SubSahara	.	.	.	.	.
OutwarOr	.	.	.	.	.
Area	.	.	.	.	.
PrScEnroll	.	.	.	.	.
LifeExp	.	.	.	.	.
GDP60	.	.	.	.	.
Mining	.	.	.	.	.
EcoOrg	.	.	.	.	.
YrsOpen	.	0.0008411755	0.0023489833	0.003459478	0.004469387
Age	.	.	.	.	.
Buddha	.	.	.	.	.
Catholic	.	.	.	.	.
Confucian	.	0.0001621431	0.007347060	0.013893165	0.019857730

Abslat	.	.	.	.	.
Spanish	.	.	.	.	.
French	.	.	.	.	.
Brit	.	.	.	.	.
WarDummy	.	.	.	.	.
LatAmerica	.	.	.	.	.
SubSahara	.	.	.	.	-0.0004079926
OutwarOr	.	.	.	.	.
Area	.	.	.	.	.
PrScEnroll	.	.	.	.	.
LifeExp	.	.	.	.	.
GDP60	.	.	.	.	.
Mining	.	.	.	.	.
EcoOrg	.	.	.	.	.
YrsOpen	0.006018196	0.006519700	0.006974666	0.007389210	0.0075017961
Age	.	.	.	.	.
Buddha	0.001442612	0.003269348	0.004934085	0.006450932	0.0078458496
Catholic	.	.	.	.	.
Confucian	0.024718431	0.028944305	0.032794347	0.036302360	0.0394825702

Abslat	.	.	.	.	.
Spanish	.	.	.	.	.
French	.	.	.	.	.
Brit	.	.	.	.	.
WarDummy	.	.	.	.	.
LatAmerica	.	.	.	.	.
SubSahara	-0.001008787	-0.001550042	-0.001756695	-0.001866927	-0.001859641
OutwarOr	.	.	.	.	.
Area	.	.	.	.	.
PrScEnroll	.	.	.	.	.
LifeExp	.	.	.	.	.
GDP60	.	.	.	.	.
Mining	.	.	.	.	.

EcoOrg	.	.	.	.	0.000116833
YrsOpen	0.007459716	0.007359030	0.007580682	0.007776759	0.007780396
Age	.	.	.	.	.
Buddha	0.009123433	0.010230306	0.010717372	0.011142999	0.011649993
Catholic	.	.	.	.	.
Confucian	0.042372076	0.045170361	0.047367782	0.049400636	0.051369519

Abslat	.	.	.	.	
Spanish	.	.	.	.	
French	.	.	.	.	
Brit	.	.	.	.	
WarDummy	.	.	.	.	
LatAmerica	.	.	-7.750287e-05	-3.738524e-04	
SubSahara	-0.0018880361	-0.0018899263	-1.908000e-03	-2.187426e-03	
OutwarOr	.	.	.	.	
Area	.	.	.	.	
PrScEnroll	.	.	.	.	
LifeExp	.	.	.	.	
GDP60	.	.	.	.	
Mining	.	.	.	.	
EcoOrg	0.0003042959	0.0004704518	6.218916e-04	7.449506e-04	
YrsOpen	0.0076243929	0.0075313623	7.530964e-03	7.433922e-03	
Age	.	.	.	-3.919301e-06	
Buddha	0.0121773288	0.0126438366	1.303751e-02	1.314536e-02	
Catholic	.	.	.	.	
Confucian	0.0532800898	0.0550396318	5.681685e-02	5.800458e-02	

Abslat	.	.	.	.	
Spanish	.	.	.	.	
French	.	.	.	.	
Brit	.	.	.	.	
WarDummy	.	.	.	.	
LatAmerica	-6.495161e-04	-9.254410e-04	-1.250017e-03	-1.750832e-03	
SubSahara	-2.465477e-03	-2.758501e-03	-3.151965e-03	-4.092164e-03	
OutwarOr	.	.	.	.	
Area	.	-5.180181e-08	-1.421854e-07	-1.977482e-07	
PrScEnroll	.	.	.	.	
LifeExp	.	.	.	.	
GDP60	.	.	.	-3.932975e-04	
Mining	.	.	6.840276e-04	3.048942e-03	
EcoOrg	8.534475e-04	9.469652e-04	1.027886e-03	1.119147e-03	
YrsOpen	7.333904e-03	7.238579e-03	7.105824e-03	7.186242e-03	
Age	-8.384239e-06	-1.341162e-05	-1.877099e-05	-2.169845e-05	
Buddha	1.320068e-02	1.312418e-02	1.296057e-02	1.257559e-02	
Catholic	.	.	.	.	
Confucian	5.902492e-02	5.974457e-02	6.024358e-02	6.024291e-02	

Abslat	.	.	
Spanish	.	.	
French	.	.	
Brit	.	.	
WarDummy	.	.	
LatAmerica	-2.049740e-03	-2.003933e-03	
SubSahara	-5.261071e-03	-5.799552e-03	
OutwarOr	.	-1.513883e-05	
Area	-2.483745e-07	-2.511687e-07	

PrScEnroll	.	.	
LifeExp	3.249307e-06	8.186014e-05	
GDP60	-1.102869e-03	-2.272266e-03	
Mining	4.601869e-03	7.482592e-03	
EcoOrg	1.253477e-03	1.324814e-03	
YrsOpen	6.986698e-03	6.950097e-03	
Age	-2.317625e-05	-2.387427e-05	
Buddha	1.209447e-02	1.181119e-02	
Catholic	-7.358827e-05	.	
Confucian	5.967612e-02	5.922153e-02	

.....
.....suppressing 59 columns and 4 rows in show(); maybe adjust options(max.print=, width
=)
.....

PrExports	.	.	.	.	.
Protestants	.	.	.	.	.
RuleofLaw	.	.	.	.	.
Popg	.	.	.	.	.
WorkPop	.	.	.	.	.
LabForce	.	.	.	.	.
HighEnroll	.	.	.	.	.
PublEduPct	.	.	.	.	.
RevnCouP	.	.	.	.	.
PolRights	.	.	.	.	.
Civllib	.	.	.	.	.
English	.	.	.	.	.
Foreign	.	.	.	.	.
RFEXDist	.	.	.	.	.
EquipInv	0.02989896	0.05202483	0.06760673	0.08042867	0.09212352
NequipInv	.	.	.	.	.
stdBMP	.	.	.	.	.
BlMktPm	.	.	.	.	.

PrExports	.	.	.	.	.
Protestants	.	.	.	.	.
RuleofLaw	.	.	.	.	.
Popg	.	.	.	.	.
WorkPop	.	.	.	.	.
LabForce	.	.	.	.	.
HighEnroll	.	.	.	.	.
PublEduPct	.	.	.	.	.
RevnCouP	.	.	.	.	.
PolRights	.	.	.	.	.
Civllib	.	.	.	.	.
English	.	.	.	.	.
Foreign	.	.	.	.	.
RFEXDist	.	.	.	.	.
EquipInv	0.1027795	0.1125887	0.1215419	0.1297117	0.1371558
NequipInv	.	.	.	.	.
stdBMP	.	.	.	.	.
BlMktPm	.	.	.	.	.

PrExports	.	.	.	.	.
Protestants	.	-8.909548e-05	-0.001614550	-2.990505e-03	-4.302230e-03

RuleofLaw	.	.	.	.	.
Popg	.	.	.	.	.
WorkPop	.	.	.	.	.
LabForce	.	.	.	.	.
HighEnroll	.	.	.	.	.
PublEdupct	.	.	.	.	.
RevnCoup	.	.	.	.	.
PolRights	.	.	.	.	.
Civllib	.	.	.	.	.
English	.	.	.	.	.
Foreign	.	.	.	.	.
RFEXDist	.	.	.	-1.181012e-06	-2.877601e-06
EquipInv	0.1504005	1.538332e-01	0.158675926	1.625835e-01	1.644492e-01
NequipInv	.	3.373210e-03	0.007744455	1.182543e-02	1.574086e-02
stdBMP	.	.	.	-3.396065e-07	-1.704587e-06
BlMktPm	.	.	.	.	.

PrExports	.	.	.	.
Protestants	-5.522139e-03	-6.607807e-03	-7.431904e-03	-8.276442e-03
RuleofLaw	.	.	.	.
Popg	.	.	.	.
WorkPop	.	.	.	.
LabForce	.	.	.	.
HighEnroll	.	.	.	.
PublEdupct	.	.	.	.
RevnCoup	.	.	.	.
PolRights	.	.	.	.
Civllib	.	.	.	.
English	.	.	.	.
Foreign	.	.	.	.
RFEXDist	-4.166059e-06	-5.562048e-06	-7.707607e-06	-8.731330e-06
EquipInv	1.655579e-01	1.666198e-01	1.673426e-01	1.683215e-01
NequipInv	1.959354e-02	2.313536e-02	2.669987e-02	2.973273e-02
stdBMP	-2.869964e-06	-3.910083e-06	-4.714721e-06	-5.498645e-06
BlMktPm	.	.	.	.

PrExports	.	.	.	.
Protestants	-9.052083e-03	-9.757811e-03	-1.039705e-02	-1.070012e-02
RuleofLaw	.	.	.	1.361109e-05
Popg	.	.	.	.
WorkPop	.	.	.	.
LabForce	.	.	.	.
HighEnroll	.	.	.	.
PublEdupct	.	.	.	.
RevnCoup	.	.	.	.
PolRights	.	.	.	.
Civllib	.	.	.	.
English	.	.	.	.
Foreign	.	.	.	.
RFEXDist	-9.587212e-06	-1.046594e-05	-1.136665e-05	-1.242972e-05
EquipInv	1.694802e-01	1.712677e-01	1.729301e-01	1.735152e-01
NequipInv	3.252305e-02	3.492974e-02	3.689608e-02	3.859131e-02
stdBMP	-6.220303e-06	-6.858693e-06	-7.463747e-06	-7.887316e-06
BlMktPm	.	.	.	.

PrExports	.	-1.787136e-04	
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Protestants	-1.094635e-02	-1.093426e-02	
RuleofLaw	1.136319e-03	1.907452e-03	
Popg	.	.	
WorkPop	.	.	
LabForce	.	.	
HighEnroll	.	.	
PublEduPct	.	.	
RevnCouP	.	.	
PolRights	.	.	
CivlLib	.	.	
English	.	.	
Foreign	.	.	
RFEXDist	-1.179680e-05	-1.113955e-05	
EquipInv	1.722068e-01	1.670267e-01	
NequipInv	4.021221e-02	4.085799e-02	
stdBMP	-7.949240e-06	-8.119496e-06	
BlMktPm	.	.	

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model3\$beta!=0

[illegible]

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```
model3$beta["PrScEnroll",]
```


s0	s1	s2	s3	s4
0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00
s5	s6	s7	s8	s9
0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00
s10	s11	s12	s13	s14
0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00
s15	s16	s17	s18	s19
0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00
s20	s21	s22	s23	s24
0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00
s25	s26	s27	s28	s29
0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00
s30	s31	s32	s33	s34
0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00
s35	s36	s37	s38	s39
1.559643e-05	7.049877e-04	1.140619e-03	2.257770e-03	3.163093e-03
s40	s41	s42	s43	s44
3.662942e-03	4.222573e-03	4.742371e-03	5.226826e-03	5.821473e-03
s45	s46	s47	s48	s49
6.379396e-03	7.175862e-03	8.114050e-03	8.900710e-03	9.654690e-03
s50	s51	s52	s53	s54
1.034464e-02	1.091332e-02	1.142913e-02	1.193047e-02	1.245326e-02
s55	s56	s57	s58	s59
1.292431e-02	1.337128e-02	1.378494e-02	1.416322e-02	1.452138e-02
s60	s61	s62	s63	s64
1.483990e-02	1.512684e-02	1.538646e-02	1.561393e-02	1.593989e-02
s65	s66	s67	s68	s69
1.620810e-02	1.653428e-02	1.683023e-02	1.709601e-02	1.734384e-02
s70	s71	s72	s73	s74
1.756329e-02	1.776771e-02	1.794096e-02	1.811346e-02	1.827023e-02
s75	s76	s77	s78	s79
1.840652e-02	1.853113e-02	1.864281e-02	1.872953e-02	1.880910e-02
s80	s81	s82	s83	s84
1.887338e-02	1.895388e-02	1.900056e-02	1.906487e-02	1.911164e-02
s85				
1.915870e-02				

[Hide](#)

```
model3$beta["GDP60",]
```

s0	s1	s2	s3
0.0000000000	0.0000000000	0.0000000000	0.0000000000
s4	s5	s6	s7
0.0000000000	0.0000000000	0.0000000000	0.0000000000
s8	s9	s10	s11
0.0000000000	0.0000000000	0.0000000000	0.0000000000
s12	s13	s14	s15
0.0000000000	0.0000000000	0.0000000000	0.0000000000
s16	s17	s18	s19
0.0000000000	0.0000000000	0.0000000000	0.0000000000
s20	s21	s22	s23
0.0000000000	0.0000000000	0.0000000000	0.0000000000
s24	s25	s26	s27
-0.0003932975	-0.0011028686	-0.0022722662	-0.0033412424
s28	s29	s30	s31
-0.0042970874	-0.0051898805	-0.0060242777	-0.0068203598
s32	s33	s34	s35
-0.0076109450	-0.0084115505	-0.0091372627	-0.0097864377
s36	s37	s38	s39
-0.0104030520	-0.0109663613	-0.0115647670	-0.0119383697
s40	s41	s42	s43
-0.0121368190	-0.0123696774	-0.0125846951	-0.0127836149
s44	s45	s46	s47
-0.0130289364	-0.0132626594	-0.0136224011	-0.0139314375
s48	s49	s50	s51
-0.0141897670	-0.0144393451	-0.0146597912	-0.0148151561
s52	s53	s54	s55
-0.0149698691	-0.0151222361	-0.0152004199	-0.0152932198
s56	s57	s58	s59
-0.0153887703	-0.0154785117	-0.0155600636	-0.0156362131
s60	s61	s62	s63
-0.0157036964	-0.0157639657	-0.0158182487	-0.0158634332
s64	s65	s66	s67
-0.0159432253	-0.0160008090	-0.0160798657	-0.0161523532
s68	s69	s70	s71
-0.0162174449	-0.0162776062	-0.0163313717	-0.0163811218
s72	s73	s74	s75
-0.0164231067	-0.0164638522	-0.0165018548	-0.0165342856
s76	s77	s78	s79
-0.0165650055	-0.0165950151	-0.0166207618	-0.0166453901
s80	s81	s82	s83
-0.0166652514	-0.0166888101	-0.0167033568	-0.0167222103
s84	s85		
-0.0167362822	-0.0167502860		